Wenxuan Wang

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Homepages: https://amazing-wx.github.io

Research interest: Motion Control of Robot, Mechanical design; Computer Vision

EDUCATION

The Chinese University of Hong Kong (CUHK) — **Hong Kong, China**

HKPFS Summer workshop in BME and MAE department — **July. 2026 – Aug.2026**

Harbin Institute of Technology, Shenzhen (HITSZ) — **Shen Zhen, China**

Master of Mechanical Engineering — **Sept. 2023 – Mar.2026**

- Supervisor: Prof. **Bing Li** (https://faculty.hitsz.edu.cn/libing)
- Relevant Courses: Numerical Analysis, Modern Control Theory, Collaborative Robotics Technology and Applications, etc.

Lanzhou University of Technology — **Lan Zhou, China**

Bachelor of Mechanical Design, Manufacturing, and Automation — **Sept. 2019 – Jun.2023**

- **GPA: 91.55**|100 Ranking **0.6%** (3/459) **National Scholarship for twice**、National Encouragement Scholarship
- Relevant Courses: Linear Algebra (100), Advanced Mathematics (95), Mechanics of Material (99), Mechanical Design (97) Electronic technology (96), Mechanical Principles (95), Calculation Method (96), Basic Mechanic Engineering Control (94)

HONORS AND AWARDS

- 100 **Outstanding National Scholarship Representatives** (only 100 a year in China) |Published in People's Daily — **May. 2022**
- **National Scholarship (Top 1, ¥8000, 0.2%)** — **2019-2020**
- **National Scholarship (Top 1, ¥8000, 0.2%)** — **2020-2021**
- **National Encouragement Scholarship (Top 10%, ¥5000)** — **2021-2022**
- The First Prize Scholarship **for three times** — **2020, 2021, 2022**
- **Provincial-Level Merit Student** (**only 2** in the same grade) — **2022**
- Merit Student / Pacemaker to Merit Student — **2020, 2021/2022**
- **Outstanding Graduates** in Lanzhou University of Technology — **2023**
- The Second Graduate Scholarship of HITSZ — **2023, 2024**

PUBLICATIONS AND ON-GONING

1. Wenxuan Wang, Yinghao Ning, Yang Zhang, Peng Xu*, Bing Li. Linear active disturbance rejection control with linear quadratic regulator for Stewart platform in active wave compensation system. ***Applied Ocean Research*.2025**. (**JCR Q1. IF=4.4---TOP**)

2. Yang Zhang, **Wenxuan Wang**, Xi Kang*, Bing Li*. Design, analysis, and experimentation of deployable multi-closed-loop truss modules for aerospace platforms. ***Mechanism and Machine Theory*, 2024 (JCR Q1. IF=5.3---TOP)**

3. Wenxuan Wang, Xiaokai Cui, Peng Xu*, Bing Li*. An improved linear quadratic regulator of shipborne Stewart platform for wave compensation, ***IEEE International Conference on Robotics and Biomimetics*. Bangkok, Thailand, 2024 (EI).**

4. Wenxuan Wang, Peng Xu*, Bing Li*. Motion control of parallel robots based on IMU and vision fusion for motion compensation *IEEE Robotics and Automation Letter*, **(JCR Q1. IF=5.3---TOP, under review)**

5. Xiaokai Cui, **Wenxuan Wang** Peng Xu*, Bing Li* Anti-disturbance Control of Joint Space of Offshore Hybrid Platform Based on Sliding Mode. ***IEEE International Conference on Robotics and Biomimetics*. Changsha, China, 2025 (EI-under review)**

RESEARCH EXPERIENCE

Research on Motion Control of a Six-DOF Parallel Robotic Stabilization Platform Based on IMU and Vision Fusion | Key number of Commissioned by China General Nuclear Power Group Laboratory| — **Dec.2023--Present**

- Designed 6-UCU parallel mechanism, completed forward/inverse kinematic solution, workspace analysis, and motion control.
- Fused position and orientation data from the fusion of IMU sensors and computer vision to enhance pose estimation accuracy
- Proposed an improved linear active disturbance rejection control (ILADRC) based on linear quadratic regulator (LQR) in joint space for each chain electric-driven actuator (PMSM) to achieve accurate stabilization of the top platform.
- Utilized LQR to replace the series PD control of the traditional LADRC to solve the problem of controller bandwidth. Used the LESO to estimate and compensate for the total system disturbance. Verified the controller's stability with the Lyapunov theory.
- Project Video: https://youtu.be/lLr3biRudsw https://www.youtube.com/watch?v=jiMR-fIs53g

Research on hybrid variable stiffness collaborative robot for abrasive machining with ILADRC algorithm| Key member of 6th China Graduate Student Robot Innovation and Design Competition **Nov. 2023 – Dec. 2024**

- Proposed a novel 3-DOF redundant parallel end-effector mechanism (2UPR-2RRU) with two rotational and one translational degrees of freedom, incorporating a lightweight structural design for enhanced performance.
- Developed an ILADRC algorithm that uses an exponential function for adaptive parameter adjustment to compensate for position errors caused by unknown disturbances, thereby significantly improved control performance.
- Designed a fuzzy-adaptive impedance control system, incorporating a fuzzy logic loop into the adaptive term to formulate a fuzzy conductance controller, effectively mitigating force overshoot during grinding force tracking.

Achievement: National Third Prize of the 6th China Graduate Student Robot Innovation and Design Competition

- Project Video: https://youtu.be/61c5QmwmCeU?si=X7ylEsiabiaAcuwA

National Training Program for Innovation and Entrepreneurship: Mechanical Design and Control of Traffic Directing Robot with 2 human-like arms |(Project Number:202210731019)----Project Leader Sept. 2022—Jun. 2023

- Designed an intelligent traffic-directing robot based on an STM32 microcontroller, equipped with dual 7-DOF humanoid arms and four Mecanum wheels for omnidirectional mobility, enabling autonomous traffic control and mobile law enforcement.
- Programmed precise servo control through PWM duty cycle modulation, allowing the robotic arms to accurately replicate standard traffic police hand signals.
- Integrated onboard cameras and image recognition algorithms with the mobile platform to identify traffic violations, addressing the limitations of fixed-location surveillance systems.
- **Achievement: National First Prize** of the 23rd China Robotics and Artificial Intelligence Competition.
  National First Prize of the International Youth Artificial Intelligence Innovation Competition.
- Project Video: https://youtu.be/FKRFKs0Cmow

Robot Perception and 3D Reconstruction - Robotics Algorithm Engineer (Intern) ***Qiaojie ShuWu*** **Jul 2025 - Sep 2025**

- Developed and integrated 3D reconstruction algorithms (VGGT, DROID-SLAM, MegaSAM, Colmap, 3DGS) using RGB videos to achieve high-precision scene reconstruction.
- Applied PGSR to extract mesh models and built simulation datasets in Isaac Sim for robot perception tasks.
- Independently debugged algorithms and documented the process to support team research
- Project Video: https://youtu.be/wCmTy1_-AXw

National Undergraduate Science and Technology Innovation Project: Motion Control of a Smart Home Security Robot and grasping | (Project Number:202110731012) ----Project Leader Oct. 2021—Jun. 2022

- Responsible for the mechanical design, simulation construction, and motion control of robotics.
- Completed the control of the robot chassis movement and the grasping action of the 5-DOF arm with Arduino chip.
- Utilized ultrasonic sensors and Arduino to communicate with each other through Usart. Adjusted the baud rate communication to achieve the obstacle avoidance function.
- **Achievement**: **National Second Prize** of the 3rd China University Intelligent Robot Creative Competition.
- Project Video: https://youtu.be/VATWh5RnMGQ

CAMPUS EXPERIENCE

Secretary of the Youth League Branch Committee | Undergraduate **Sept. 2019- Jun. 2023**

- Organized and planned **30+** class league activities, achieving a participation rate of over **95%**, effectively enhancing class cohesion. Led the class in school-wide and community volunteer activities, with 50+ students participation.

COMPETITIONS AND AWARDS

- 1 **National First Prize** of the 23rd China Robotics and Artificial Intelligence Competition. **Dec.2021**
- 2 **National First Prize** of the International Youth Artificial Intelligence Innovation Competition. **Nov.2021**
- 3 **National First Prize** of the 14th National 3D Digital Innovation Design Competition. **Aug.2021**
- 4 **National Second Prize** of the 12th Process Equipment Practice and Innovation Competition. **Dec.2021**
- 5 **National Second Prize** of the 17th Embedded Artificial Intelligence Competition for Students **Dec.2021**
- 6 **National Second Prize** of the 3rd China University Intelligent Robot Creative Competition. **Dec.2020**
- 7 **Bronze Award** of the 7th China College Students "Internet+" Innovation and Entrepreneurship Competition **Dec.2021**
- 8 **First Prize** at the provincial level of the Chinese Mathematics Competitions **Dec.2020**
- 9 **First Prize** at the provincial level of the China Undergraduate Mathematical Contest in Modeling **Nov.2020**
- 10 **Honorable Mention** of Mathematical Contest in Modeling **Apr.2021**